**Supplementary Table 1: Comparison of models by predictive performance.** Mean AUCs for first-line drugs and mean AUCs for second-line drugs are compared between logistic regression + L2 regularization and the MD-CNN (LR v MD-CNN), logistic regression + L2 regularization and the WDNN (LR v WDNN), and so on. Welch's two-sample, two-sided t-test is used to determine the statistical significance of the difference in mean AUCs between models. Benjamini-Hochberg p-value correction is performed for multiple testing. Note that ethionamide, and levofloxacin are excluded from the list of second-line drugs because the original WDNN was not trained on these drugs. LR = logistic regression + L2 regularization; WDNN = wide and deep neural network; MD-CNN = multi-drug convolutional neural network; SD-CNN = single-drug convolutional neural network.

| Model comparison | Drug group | Model 1 | Model 1 AUC | Model 2 | Model 2 AUC | Difference in mean AUCs [95% CI] | p | Benjamini-Hochberg p |
|---|---|---|---|---|---|---|---|---|
| **LR v MD-CNN** | First-line drugs | LR | 0.923 | MD-CNN | 0.948 | -0.026 [-0.047, -0.0039] | 0.032 | 0.059 |
| **LR v WDNN** | First-line drugs | LR | 0.923 | WDNN | 0.960 | -0.037 [-0.058, -0.016] | 0.0022 | 0.0054 |
| **LR v SD-CNN** | First-line drugs | LR | 0.923 | SD-CNN | 0.938 | -0.015 [-0.037, 0.006] | 0.18 | 0.18 |
| **WDNN v MD-CNN** | First-line drugs | WDNN | 0.960 | MD-CNN | 0.948 | 0.011 [-0.0015, 0.024] | 0.10 | 0.13 |
| **WDNN v SD-CNN** | First-line drugs | WDNN | 0.960 | SD-CNN | 0.938 | 0.022 [0.0096, 0.034] | 0.0014 | 0.0042 |
| **MD-CNN v SD-CNN** | First-line drugs | MD-CNN | 0.948 | SD-CNN | 0.938 | 0.010 [-0.0034, 0.024] | 0.16 | 0.18 |
| **LR v MD-CNN** | Second-line drugs | LR | 0.860 | MD-CNN | 0.912 | -0.051 [-0.081, -0.022] | 0.0010 | 0.0042 |
| **LR v WDNN** | Second-line drugs | LR | 0.860 | WDNN | 0.924 | -0.064 [-0.092, -0.036] | 0.000044 | 0.00053 |
| **LR v SD-CNN** | Second-line drugs | LR | 0.860 | SD-CNN | 0.888 | -0.028 [-0.058, 0.0028] | 0.082 | 0.12 |
| **WDNN v MD-CNN** | Second-line drugs | WDNN | 0.924 | MD-CNN | 0.912 | 0.013 [-0.0055, 0.031] | 0.18 | 0.18 |
| **WDNN v SD-CNN** | Second-line drugs | WDNN | 0.924 | SD-CNN | 0.888 | 0.036 [0.016, 0.057] | 0.00083 | 0.0042 |
| **MD-CNN v SD-CNN** | Second-line drugs | MD-CNN | 0.912 | SD-CNN | 0.888 | 0.024 [0.0023, 0.045] | 0.034 | 0.059 |

**Supplementary Table 2: Comparison of models by predictive performance**, measured as mean sensitivity and mean specificity for logistic regression + L2 regularization, WDNN (wide and deep neural network), MD-CNN (multi-drug convolutional neural network), and SD-CNN (single-drug convolutional neural network). For each drug, highest sensitivity and specificity values are emboldened. The WDNN was not initially trained on levofloxacin or ethionamide and thus it is not evaluated for these drugs.

| | **Logistic Regression** | | **WDNN** | | **MD-CNN** | | **SD-CNN** | |
|---|---|---|---|---|---|---|---|---|
| **Drug** | **Sensitivity, %** | **Specificity, %** | **Sensitivity, %** | **Specificity, %** | **Sensitivity, %** | **Specificity, %** | **Sensitivity, %** | **Specificity, %** |
| Rifampicin | 90.9 | 95.5 | 92.6 | 96.2 | **96.9** | 94.4 | 93.1 | **97.2** |
| Isoniazid | 90.6 | 95.7 | 92.0 | 97.0 | **96.4** | 93.6 | 91.8 | **97.0** |
| Pyrazinamide | 79.0 | 87.7 | 87.8 | 90.5 | **88.5** | 89.3 | 87.6 | **95.1** |
| Ethambutol | 85.1 | 84.4 | **90.6** | 85.5 | 86.1 | **91.9** | 87.5 | 87.3 |
| Streptomycin | 85.1 | 85.7 | 89.0 | 85.5 | **89.7** | 87.4 | 88.3 | **89.2** |
| Levofloxacin | 87.2 | 84.0 | - | - | **97.4** | 90.9 | 90.9 | **94.2** |
| Capreomycin | 72.3 | 88.4 | 78.1 | 88.1 | **89.1** | 80.3 | 70.8 | **95.1** |
| Amikacin | 68.6 | 96.1 | 74.1 | 95.6 | **93.4** | 82.1 | 72.4 | **96.4** |
| Moxifloxacin | 71.7 | 84.0 | 71.8 | 90.2 | **87.9** | 78.9 | 67.8 | **95.0** |
| Ofloxacin | 68.6 | 94.3 | 78.3 | 93.7 | **92.3** | 84.8 | 77.5 | **96.4** |
| Kanamycin | 77.2 | 94.9 | 83.9 | 95.5 | **93.9** | 88.3 | 84.4 | **97.2** |
| Ethionamide | 70.8 | 77.6 | - | - | **77.2** | **81.7** | 74.1 | 77.7 |
| Ciprofloxacin | 97.5 | 91.0 | 98.3 | 97.1 | **99.2** | **98.3** | 94.9 | 98.2 |

**Supplementary Table 3: Comparison of the MD-CNN model with the WHO catalog**, using mean sensitivity and mean specificity for the MD-CNN (multi-drug convolutional neural network), and the WHO (World Health Organization) catalog for each drug. The WHO catalog has no information on ofloxacin and ciprofloxacin, hence their exclusion.

| | **MD-CNN** | | **WHO catalog** | |
|---|---|---|---|---|
| **Drug** | **Sensitivity, %** | **Specificity, %** | **Sensitivity, %** | **Specificity, %** |
| Rifampicin | 96.9 | 94.4 | 93.2 | 96.2 |
| Isoniazid | 96.4 | 93.6 | 87.8 | 97.4 |
| Pyrazinamide | 88.5 | 89.3 | 61.7 | 96.5 |
| Ethambutol | 86.1 | 91.9 | 78.8 | 89.1 |
| Streptomycin | 89.7 | 87.4 | 64.9 | 95.6 |
| Levofloxacin | 97.4 | 90.9 | 92.3 | 94.2 |
| Capreomycin | 89.1 | 80.3 | 67.8 | 96.0 |
| Amikacin | 93.4 | 82.1 | 71.2 | 97.5 |
| Moxifloxacin | 87.9 | 78.9 | 66.2 | 95.1 |
| Kanamycin | 93.9 | 88.3 | 83.5 | 97.2 |
| Ethionamide | 77.2 | 81.7 | 65.5 | 79.5 |

**Supplementary Table 4: CNN models generalize on new data with realistic resistance proportions** as measured by sensitivity and specificity of the CNN models' resistance predictions based on genomic data from the CRyPTIC database (see **Methods**) not otherwise used during their training and cross-validation. Sensitivity and specificity are calculated for both complete data and data down-sampled to limit the proportion of rifampicin-resistant tuberculosis strains to 5%, their estimated overall global prevalence. Down-sampling is performed 10 times, for which mean resistance numbers and proportions are calculated. The drugs below represent those antibiotics for which phenotype information is available in the CRyPTIC database. MD-CNN = multi-drug convolutional neural network; SD-CNN = single-drug convolutional neural network; 95% CI = 95% confidence interval.

| | **MD-CNN complete data** | | | | **MD-CNN down-sampled data** | | | |
|---|---|---|---|---|---|---|---|---|
| **Drug** | **Resistant (n)** | **Proportion** | **Sensitivity, % [95% CI]** | **Specificity, % [95% CI]** | **Resistant (mean n)** | **Proportion (mean)** | **Sensitivity, % [95% CI]** | **Specificity, % [95% CI]** |
| Rifampicin | 2791 | 0.402 | 97.6 [97.6-97.7] | 95.3 [95.3-95.3] | 219 | 0.062 | 97.7 [97.5-97.9] | 97.7 [97.5-97.9] |
| Isoniazid | 3556 | 0.489 | 96.0 [95.9-96.0] | 94.5 [94.5-94.6] | 191.5 | 0.052 | 97.8 [97.6-98.0] | 97.0 [96.8-97.2] |
| Ethambutol | 1170 | 0.234 | 95.6 [95.6-95.7] | 88.3 [88.2-88.4] | 83.8 | 0.030 | 97.2 [96.9-97.5] | 97.6 [97.3-97.9] |
| Levofloxacin | 973 | 0.159 | 82.1 [82.0-82.3] | 97.8 [97.7-97.9] | 64.2 | 0.020 | 84.8 [84.0-85.7] | 99.0 [98.1-99.8] |
| Amikacin | 431 | 0.607 | 85.4 [85.2-85.6] | 90.3 [90.1-90.5] | 29.5 | 0.008 | 88.7 [87.7-89.8] | 98.3 [97.2-99.3] |
| Moxifloxacin | 794 | 0.147 | 91.4 [91.3-91.5] | 92.4 [92.3-92.6] | 55.0 | 0.020 | 95.1 [94.5-95.6] | 98.6 [98.0-99.2] |
| Kanamycin | 550 | 0.076 | 80.0 [79.9-80.2] | 91.2 [91.1-91.4] | 36.9 | 0.010 | 83.6 [82.5-84.8] | 98.5 [97.3-99.6] |
| Ethionamide | 929 | 0.147 | 77.5 [77.3-77.6] | 91.4 [91.3-91.6] | 53.9 | 0.015 | 80.9 [79.8-82.0] | 98.1 [97.0-99.2] |

| | **SD-CNN complete data** | | | | **SD-CNN down-sampled data** | | | |
|---|---|---|---|---|---|---|---|---|
| **Drug** | **Resistant (n)** | **Proportion** | **Sensitivity, % [95% CI]** | **Specificity, % [95% CI]** | **Resistant (mean n)** | **Proportion (mean)** | **Sensitivity, % [95% CI]** | **Specificity, % [95% CI]** |
| Rifampicin | 2791 | 0.402 | 96.6 [96.6-96.6] | 97.1 [97.1-97.2] | 219 | 0.062 | 96.8 [96.5-97.0] | 98.6 [98.4-98.8] |
| Isoniazid | 3556 | 0.489 | 94.8 [94.7-94.8] | 98.0 [98.0-98.1] | 191.5 | 0.052 | 96.0 [95.7-96.2] | 98.5 [98.2-98.8] |
| Ethambutol | 1170 | 0.234 | 94.4 [94.4-94.5] | 89.5 [89.4-89.5] | 83.0 | 0.029 | 95.9 [95.5-96.3] | 95.5 [95.1-95.9] |
| Levofloxacin | 973 | 0.159 | 89.1 [89.0-89.2] | 97.9 [97.8-98.0] | 64.6 | 0.020 | 92.0 [91.4-92.6] | 99.3 [98.7-99.9] |
| Amikacin | 431 | 0.607 | 85.8 [85.6-85.9] | 97.3 [97.1-97.4] | 29.6 | 0.008 | 89.3 [88.3-90.3] | 98.8 [97.9-99.8] |
| Moxifloxacin | 794 | 0.147 | 93.6 [93.6-93.7] | 94.9 [94.8-95.0] | 54.9 | 0.020 | 96.3 [95.9-96.8] | 98.8 [98.3-99.3] |
| Kanamycin | 550 | 0.076 | 84.7 [84.5-84.8] | 97.7 [97.6-97.9] | 37.2 | 0.010 | 89.0 [88.0-89.9] | 99.5 [98.5-100.0] |
| Ethionamide | 929 | 0.147 | 73.7 [73.6-73.8] | 96.8 [96.6-96.9] | 52.7 | 0.015 | 71.7 [70.6-72.8] | 98.1 [97.0-99.2] |

**Supplementary Table 5: High saliency sites according to DeepLIFT reflect a high proportion of known resistance-conferring variants.** Percent of high saliency sites for each SD-CNN (single-drug convolutional neural network) model that are classified as known to be involved in antibiotic resistance.

| *drug* | *percent known, top 1%* | *percent known, top 0.1%* |
|---|---|---|
| *AMIKACIN* | 11.5% | 75.0% |
| *CAPREOMYCIN* | 16.9% | 57.1% |
| *ETHAMBUTOL* | 8.3% | 61.5% |
| *ISONIAZID* | 16.1% | 44.4% |
| *KANAMYCIN* | 14.9% | 87.5% |
| *LEVOFLOXACIN* | 6.3% | 75.0% |
| *MOXIFLOXACIN* | 12.5% | 100.0% |
| *OFLOXACIN* | 10.4% | 100.0% |
| *PYRAZINAMIDE* | 85.4% | 90.0% |
| *RIFAMPICIN* | 57.0% | 100.0% |
| *STREPTOMYCIN* | 36.4% | 85.7% |

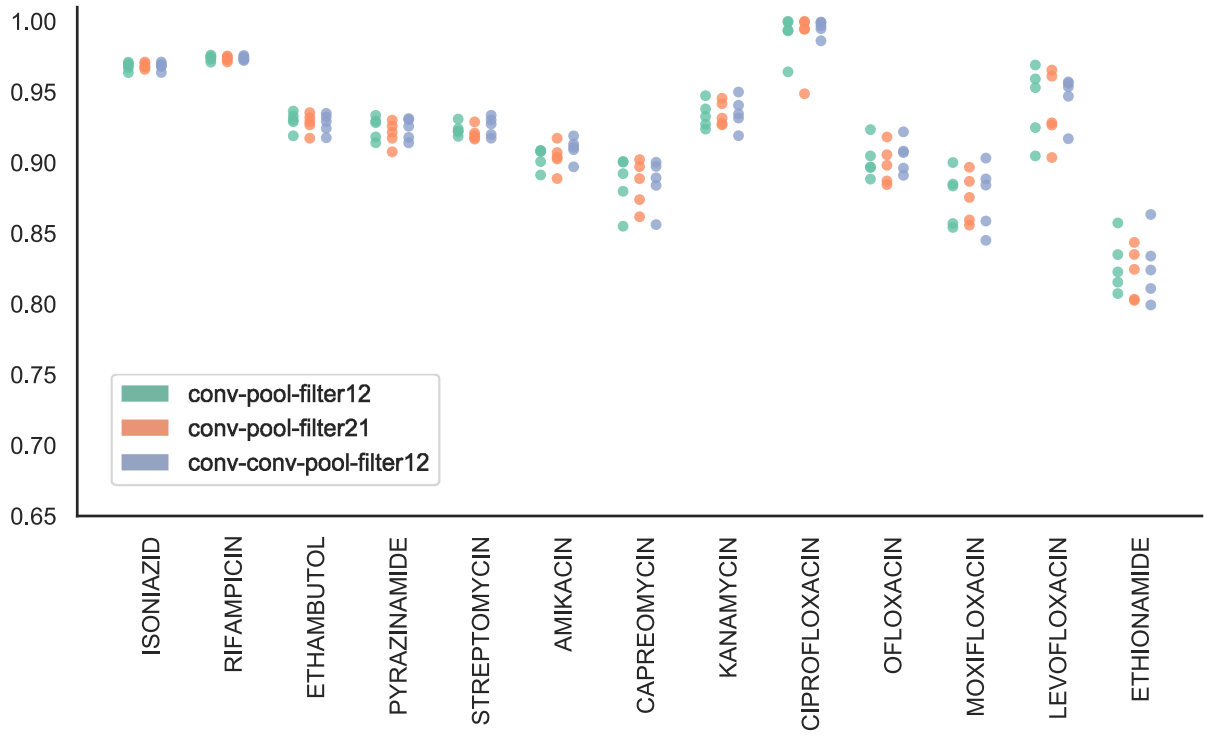
**Supplementary Table 6: Newly discovered sites associated with antibiotic resistance.** Sites in the top 0.1% of saliency scores for any drug that are not designated as known to cause resistance in the WHO catalog.

| position | locus | gene | Max saliency score | drug |
|---|---|---|---|---|
| 7585 | gyrBA | gyrA | 0.44793767 | LEVOFLOXACIN |
| 1472337 | rrs-rrl | rrs | 0.33726892 | AMIKACIN |
| 1472358 | rrs-rrl | rrs | 4.3074465 | STREPTOMYCIN |
| 1472358 | rrs-rrl | rrs | 1.2354082 | CAPREOMYCIN |
| 1474001 | rrs-rrl | rrl | 1.4282935 | CAPREOMYCIN |
| 1474001 | rrs-rrl | rrl | 0.18678115 | AMIKACIN |
| 1476056 | rrs-rrl | rrl | 0.75932384 | CAPREOMYCIN |
| 1673423 | FabG1-inhA | intergenic | 4.7082443 | ISONIAZID |
| 2288944 | pncA | pncA | 6.616614 | PYRAZINAMIDE |
| 2715338 | eis | intergenic | 2.4716604 | KANAMYCIN |
| 2726139 | oxyR-ahpC | intergenic | 5.018302 | ISONIAZID |
| 2726141 | oxyR-ahpC | intergenic | 4.3265367 | ISONIAZID |
| 2726142 | oxyR-ahpC | intergenic | 4.2414837 | ISONIAZID |
| 2726145 | oxyR-ahpC | intergenic | 4.2477694 | ISONIAZID |
| 4243217 | embCAB | intergenic | 2.8458147 | ETHAMBUTOL |
| 4243222 | embCAB | intergenic | 3.7729878 | ETHAMBUTOL |
| 4243225 | embCAB | intergenic | 2.9434128 | ETHAMBUTOL |
| 4247402 | embCAB | embB | 2.7454894 | ETHAMBUTOL |
| 4249518 | embCAB | embB | 2.9824004 | ETHAMBUTOL |
| 4327484 | ethAR | intergenic | 2.4312048 | ETHIONAMIDE |

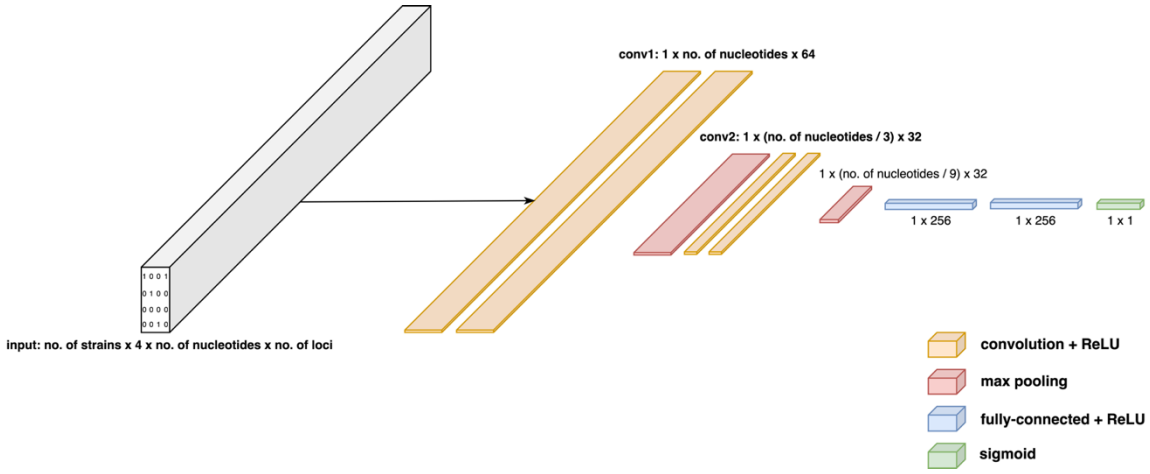
**Supplementary Table 7: Lineage-defining variants represent a small fraction of high saliency positions in the SD-CNN (single-drug convolutional neural network) models.** The number of sites per SD-CNN model, the number of those sites that have been mutated (SNP or INDEL) at least once, the number of sites that are correlated with lineage-defining variants **(Methods)**, the number of hits (sites that comprise the top 0.1% of high-saliency positions), and the number of hit sites that are correlated with lineage are shown.

| *drug* | *N sites* | *N sites mutated* | *N lineage sites* | *N hits* | *N lineage hits* | *% lineage sites* | *% lineage hits* |
|---|---|---|---|---|---|---|---|
| *AMIKACIN* | 12168 | 3997 | 3 | 121 | 2 | 0.08% | 1.65% |
| *CAPREOMYCIN* | 12168 | 3384 | 2 | 121 | 1 | 0.06% | 0.83% |
| *ETHAMBUTOL* | 20582 | 3468 | 19 | 205 | 11 | 0.55% | 5.37% |
| *ISONIAZID* | 14940 | 6161 | 15 | 149 | 5 | 0.24% | 3.36% |
| *KANAMYCIN* | 12168 | 3997 | 3 | 121 | 1 | 0.08% | 0.83% |
| *LEVOFLOXACIN* | 4859 | 1410 | 8 | 48 | 1 | 0.57% | 2.08% |
| *MOXIFLOXACIN* | 4859 | 1410 | 8 | 48 | 3 | 0.57% | 6.25% |
| *OFLOXACIN* | 4859 | 1410 | 8 | 48 | 4 | 0.57% | 8.33% |
| *PYRAZINAMIDE* | 17864 | 4820 | 10 | 178 | 4 | 0.21% | 2.25% |
| *RIFAMPICIN* | 7910 | 1955 | 9 | 79 | 4 | 0.46% | 5.06% |
| *STREPTOMYCIN* | 18252 | 4258 | 3 | 182 | 3 | 0.07% | 1.65% |

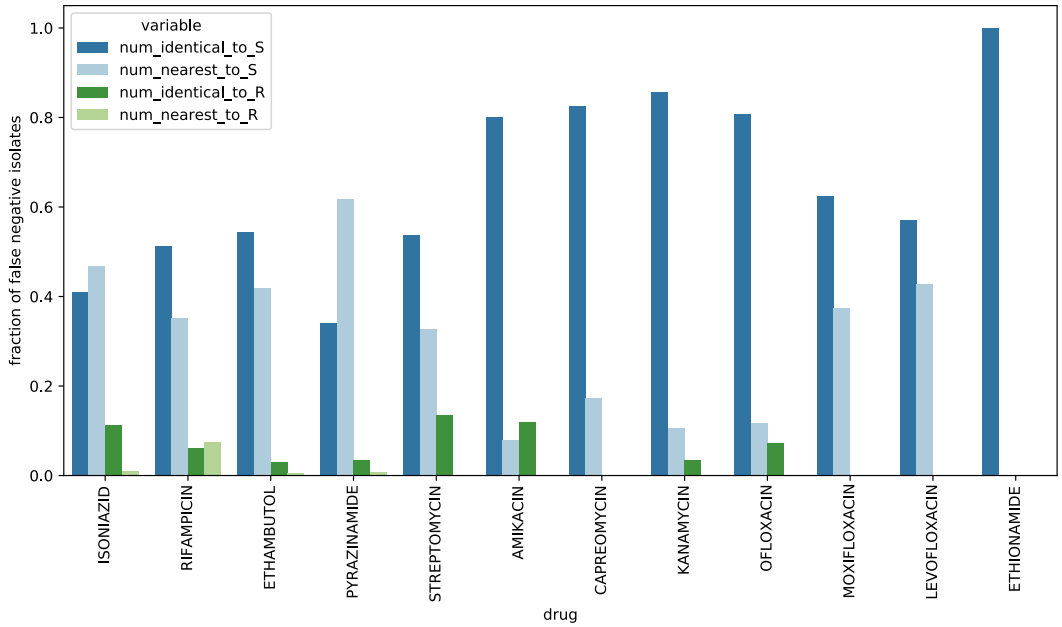
**Supplementary Figure 1: Comparison of model architectures for MD-CNN.** Data are presented as AUCs from five-fold cross-validation for each drug, with results for each of the three multi-drug convolutional neural network architectures. We perform five-fold cross validation using our training data (N=10,201 isolates) for three architectures: a 2x conv-conv-pool architecture with filter size 12, a 2x conv-pool architecture with filter size 12, and 2x conv-pool architecture with filter size 21. The 2x conv-conv-pool architecture was chosen for further study.



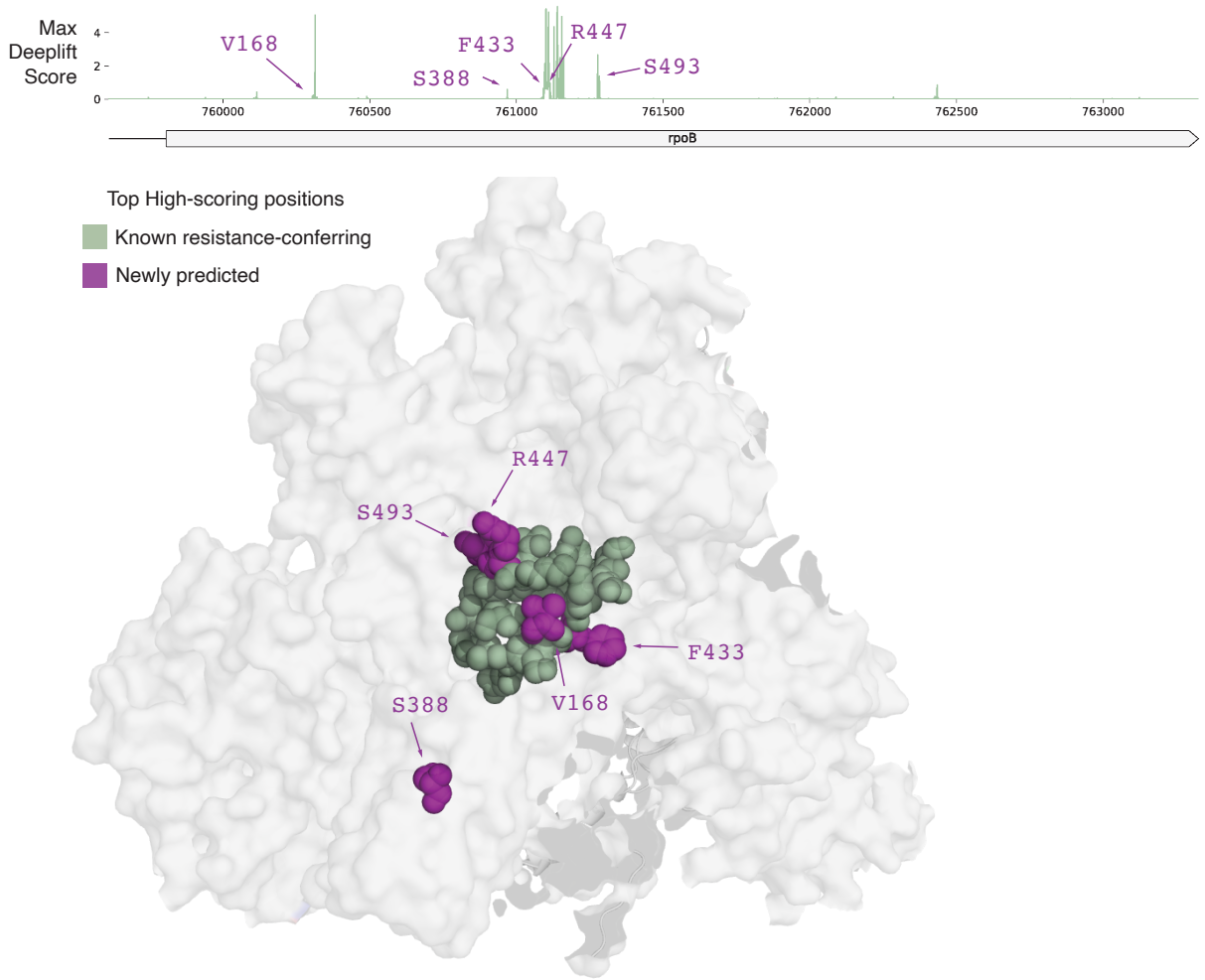
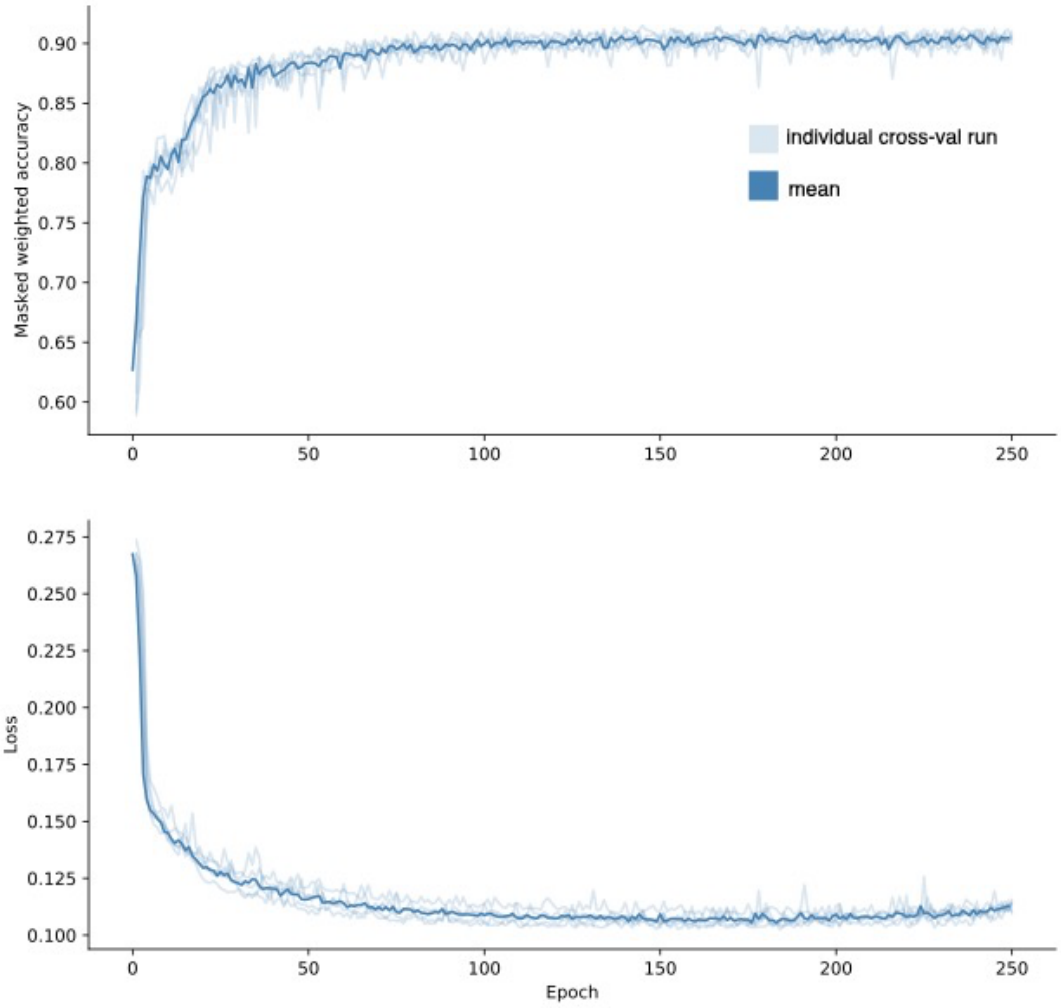
**Supplementary Figure 2:** schematic diagram of the single-drug convolutional neural network (SD-CNN). In the output layer, there is one node to compute a probability of resistance for the respective anti-TB drug. The input consists of '10,201' isolates (TB strains) for which there is resistance phenotype data for at least 2 anti-TB drugs; 4 for one-hot encoding of each nucleotide; no. of nucleotides in the locus/loci of interest for the respective drug (selected as detailed in **Methods**).



**Supplementary Figure 3: Comparison of false negatives with training data phenotypes for SD-CNN.** Proportion of false negatives that are nearest to resistant or sensitive isolates in the training dataset. For each falsely predicted isolate, we calculate the consensus phenotype (majority R or majority S) for the nearest isolate(s) in the training dataset. We differentiate between identical isolates (isolates with 0 mutational distance) and nearest non-identical isolates. R = resistant; S = sensitive.



**Supplementary Figure 4: Rifampicin saliency score plot for SD-CNN model.** Plot of maximum DeepLIFT saliency scores across the rpoB gene (note that both *rpoB* and *rpoC* were included as inputs, only *rpoB* is shown here for resolution). The five highest saliency scores not previously known to cause resistance are highlighted in purple and visualized on the rpoB protein structure (PDB ID 5UH5) along with all higher-scoring known variants.



**Supplementary Figure 5: MD-CNN validation accuracy and loss during cross-validation.** Five-fold cross validation was performed on the MD-CNN to select the optimal number of epochs for training. The mean validation accuracy and validation loss are computed across the five splits. Final number of epochs selected is 150.